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The role of artificial intelligence in diabetic retinopathy screening in type 1 diabetes: A systematic review

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ABSTRACT

Background/objectives: Diabetic retinopathy (DR) is one of the leading causes of blindness in adults worldwide and represents a critical complication in both type 1 (T1D) and type 2 (T2D) diabetes. Artificial Intelligence (AI) offers a promising opportunity to enhance both the accuracy of screening and the efficiency of ongoing care management, assisting healthcare providers in mitigating the incidence and complications of DR.
Methods: Systematic review of the literature was conducted following PRISMA guidelines. Searches were performed using PubMed-Medline, Scopus, and Embase databases, with the protocol registered on the Open Science Framework (OSF) database: (doi.org/10.17605/OSF.IO/TJ9UH). A predefined search strategy utilizing Boolean operators was applied, and two researchers independently selected articles, with a third resolving any discrepancies.
Results: Of the 2127 articles identified, 8 studies were included. The results highlighted that AI is particularly effective in enhancing the DR screening process in patients with T1D, offering rapid and reliable analysis. Healthcare providers reported positive feedback, noting its significant contribution to improving patient management.
Conclusions: The integration of AI into DR care pathways shows substantial potential for improving early diagnosis and disease management, particularly for patients with T1D. Further research is required to optimize AI implementation and ensure its positive and sustainable impact on public health.

1. Introduction

Non-communicable diseases (NCDs) represent one of the main challenges for global healthcare systems. These include heart disease, stroke, cancer, diabetes, and chronic lung diseases, which, according to official estimates from the World Health Organization (WHO), are responsible for 74 % of all deaths worldwide.[1] NCDs affect individuals of all ages and geographical backgrounds, and even in low- and middle-income countries, where they cause about 31.4 million deaths, they are beginning to have a significant impact on the healthcare systems involved.[2] Among these, diabetes stands out as one of the most common and complex conditions to manage for care providers,[3,4] both due to its impact on the quality of life of patients[5] and the costs associated with its extended- and medium-term management.[6] According to the International Diabetes Federation (IDF), it is estimated that today, 537 million adults live with this disease, a 16 % increase from the 2019 estimates.[7]

This is mainly due to the ageing of the general population, as well as improvements in prognosis and associated comorbidities.[8] One of the main neurovascular complications of diabetes is diabetic retinopathy (DR), which is the leading cause of blindness among working-age adults,

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according to the Centers for Disease Control and Prevention (CDC).[9] Currently, 93 million individuals worldwide are affected, and it is estimated that the number will reach 16 million in the Americas by 2050.[10] Several studies have shown that affected individuals are prone to depression, loss of confidence, and other adverse emotional reactions, further complicating the management of their overall well-being.[11]

Complementary care strategies could alleviate emotional symptoms, especially in a vulnerable population such as those with diabetes.[12] DR develops as a result of damage to the retina's small blood vessels caused by prolonged hyperglycemia, which can thicken and cause the loss of the ability to regulate blood flow, leading to swelling and loss of visual function.[13] Compared to adults with type 2 diabetes (T2D), those with type 1 diabetes (T1D) are at a higher risk of developing complications from DR due to the early onset of the disease and the inherent blood sugar variability associated with T1D.[14] Moreover, the longer duration of the disease in T1D patients further increases their cumulative risk for developing advanced stages of DR. These characteristics make T1D patients a particularly vulnerable group that warrants targeted screening and prevention strategies. Despite this, most current research and AI-based screening tools have focused predominantly on T2D populations, leaving a critical gap in evidence and clinical tools tailored for T1D patients.[7,9,10,15] This increased susceptibility to complications such as DR highlights the need to pay special attention to individuals with T1D in order to manage early prevention strategies and minimize the risk of DR-related complications. Moreover, prevention and early diagnosis are essential to avoid the progression of the disease to more severe stages, such as diabetic macular edema or proliferative retinopathy.[16,17] Studies have shown that preventive interventions can reduce the risk of vision loss by up to 57 %.[17] However, a shift from tertiary to primary and secondary prevention measures is urgently needed, particularly in low- and middle-income countries.[18]

In this context, artificial intelligence (AI) is poised to play a transformative role in medical diagnostics, screening, prognosis, and decision support for the management of DR.[19] AI enables large-scale screening, personalized predictive models, and improved cost-effectiveness in care delivery.[20] This technology has also been applied to other chronic conditions, including cardiovascular, rheumatic, renal, intestinal, respiratory, psychiatric diseases, and cancer.[21–29] AI systems leverage machine learning (ML), and deep learning (DL) to autonomously improve their performance and process complex datasets.[30] Many AI's applications function as augmented intelligence, combining human decision-making capabilities (e.g., common sense, morality, and imagination) with AI-specific functions (e.g., pattern recognition and machine learning) to achieve high clinical accuracy. For DR, fully automated retinal image classification could enable healthcare providers to efficiently decide whether a patient requires referral to an ophthalmologist or additional screening, thereby reducing the overall burden of diabetes on healthcare systems.[31]

Previous systematic reviews[32,33] have studied the potential benefits of adopting AI for the prevention of retinopathy complications in diabetic individuals, but to date, none of these have focused specifically on the cohort of patients affected by T1D.

2. Materials and methods

2.1. Review design

To ensure methodological rigor and the relevance of the selected studies, a preliminary research protocol was developed as part of the Systematic Review (SR). The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were adopted for reporting purposes (Supplementary file 1),[34,35] and the Cochrane Handbook for SRs was consulted during the preliminary phase.[36]

2.2. Protocol registration

The protocol for this SR was registered in the Open Science Framework (OSF) Register: doi.org/10.17605/OSF.IO/TJ9UH.[37]

2.3. Study objectives and research question

The primary objective of this study was to assess the effectiveness of AI in screening for DR in individuals with T1D, focusing on sensitivity, specificity, and accuracy compared to conventional screening methods. Additionally, both quantitative and qualitative results were analyzed. The review aimed to address the following research questions:

Primary research question:

What is the effectiveness of AI in screening for DR in individuals with T1D?

Secondary research questions:

- *What is the impact of AI on the early identification of DR complications, such as diabetic macular edema and proliferative retinopathy?*
- *How do different AI systems based on ML and DL compare in terms of performance?*
- *Considering technological, economic, or ethical factors, what are the limitations and barriers to implementing AI in DR screening?*
- *How can AI be integrated into existing healthcare systems to support managing and following up patients with DR?*
- *How can AI be compared to conventional methods for DR screening in terms of sensitivity, specificity, and accuracy?*

The research questions for this review were formulated using the PICOS framework[38] (Population, Intervention, Comparison, Outcome, Study design), specifically P, individuals with T1D at risk of developing DR; I, DR screening using AI-based systems; C, conventional screening or no comparison; O, qualitative/quantitative outcomes; and S, primary studies.

2.4. Search strategy

Searches were conducted in the PubMed/Medline, Scopus, and Embase databases, update of the date October 31, 2024. The search strategy included keywords such as "Artificial Intelligence," "Diabetic Retinopathy," and "Diabetes Mellitus, Type 1." Boolean operators were employed to develop tailored search strategies for each database (Supplementary file 2). The free version of Mendeley citation management software (Version 2.120.0)[39] was used to compile the screening database and remove duplicates. To minimize selection bias, articles were manually reviewed, and the bibliographies of the included studies were thoroughly examined.

2.5. Inclusion and exclusion criteria

The inclusion criteria encompassed primary studies, primarily in English, published in the last 5 years, relevant to the study's objective and involving adult or young individuals with T1D undergoing DR screening. The exclusion criteria eliminated all other types of studies (e.g., editorials, comments, reviews, and protocol studies) and typically pediatric population (under 14 years old in mean age of sample considered in the study included), with manuscript written in Chinese, were excluded.

2.6. Quality assessment and risk of bias evaluation

The risk of bias and methodological quality of the included studies were independently assessed by two researchers (X.X. and X.X.) using a double-blind methodology. Any disagreements were resolved by a third impartial researcher (X.X.). The risk of bias and the methodological quality of the included studies were assessed using the Critical Appraisal

Table 1
Summary of included studies.

Author and year	Country	Study design	Sample (patients with T1D)	Mean age and gender distribution	Timing study and setting	Type of AI	Sensitivity, specificity and accuracy	Specific objective	Results
Wolf et al.[42] 2024	USA	RCT	EG (n = 81) (T1D, n = 59) CG (n = 82) (T1D, n = 12)	15.2 years; 58 % female	24 Nov 2021–6 Dec 2022 Clinical center	Autonomous AI system	85.7 % sensitivity and 79.3 % specificity	Evaluate the impact of AI on participation in DR screening.	AI significantly improved the completion rate of DED compared to the CG.
Montaser et al.[43] 2024	USA	Retrospective explorative study	N = 60	21.2 years (DR)/41.8 years (control); 60 % female (DR)/50 % female (control)	Up to 7 years retrospective Clinical center	ML (SVM, LDA, RF)	Not clearly specified	Classify adults with T1D with or without incident DR using CGM data.	ML models with CGM data predicted the risk of DR. A linear SVM model achieved an AUC-ROC of 0.90.
Lee et al.[44] 2022	USA	Non-randomized prospective study	N = 265 (T1D, n = 86)	50 years ±17; 45 % female	Ongoing Remote	DL	81.0 % sensitivity and 73.5 % specificity 82.8 % accuracy	Evaluate the diagnostic accuracy of a DL model for DR using images acquired with the Optomap device.	The DL model achieved good sensitivity, specificity, and AUC-ROC.
Mathenge et al.[45] 2022	Rwanda	RCT Multicenter	EG (n = 136) (T1D, n = 63) CG (n = 139) (T1D, n = 56)	50.7 years; 58.2 % female	Mar 2021–Jul 2021 Clinical center	DL	Not clearly specified	Evaluate the impact of AI on participation in DR screening.	AI has improved participation in screening. Male gender and age were associated with a higher likelihood of DR.
Al-Sari et al.[46] 2022	Denmark	Retrospective cohort study	N = 383	54.8 years; 44.6 % female	Median 5.4 years Clinical center	ML	Not clearly specified	Develop risk prediction models for microvascular complications (including DR) in T1D.	ML models with clinical and omics data predicted the progression of complications. Eight biomolecules showed strong predictive value.
Lo et al.[47] 2021	Taiwan	Experimental study	Retinal images; subjects with T1D EG_1 (n = 8408) EG_2 (n = 7064; T1D = 475)	55.4 years; not specified	2013–2020 Hospital	DL	75.75 % average sensitivity and 80.5 average sensibility for the 4 models tested	Evaluate the effect of data homogeneity in the training of deep learning models for DR.	Models trained with data from a single center. T1D subjects showed minor performance when tested with external data.
Ipp et al.[48] 2021	USA	Multicenter cross-sectional diagnostic study	N = 893 (T1D, n = 206)	53.9 years; approximately equal	17 Apr 2017–30 May 2018 Several clinical center: primary care (6), general ophthalmology (6), and retina specialty (3) centers.	Autonomous AI system (mtmDR and vtDR).	mtmDR (sensitivity: 95.5 %; 95 % CI, 92.4 %–98.5 % and specificity: 85.0 %; 95 % CI, 82.6 %–87.4 %) and vtDR (sensitivity: 95.1 %; 95 % CI, 90.1 %–100 % and specificity: 89.0 %; 95 % CI, 87.0 %–91.1 %) without	Evaluate the sensitivity and specificity of an AI system for the autonomous detection of DR.	The AI system demonstrated good sensitivity and specificity for the identification of mtmDR and vtDR. The data were collected in 15 care centers, including primary care centers, general ophthalmology, and retina specialists.
Scheetz et al.[49] 2021	Australia	Mixed Methods Approach: Prospective Observational	N = 203 (T1D, n = 70)	56 years; 50.2 % male Clinical center	Mar 2018–May 2019	Autonomous AI system	96.9 % sensitivity 87.7 % specificity	Evaluate the diagnostic accuracy of an AI system for DR	The AI showed good sensitivity and specificity in detecting DR. The diagnostic

(continued on next page)

Table 1 (*continued*)

Author and year	Country	Study design	Sample (patients with T1D)	Mean age and gender distribution	Timing study and setting	Type of AI	Sensitivity, specificity and accuracy	Specific objective	Results
		Study and Interviews						in a real-world setting.	performance was better in patients with T1D compared to those with T2D.

Legend. EG: experimental group; CG: control group; RCT: randomized controlled trial; AI: artificial intelligence; DR: diabetic retinopathy; DED: diabetic eye exams; ML: machine learning; DL: deep learning; T1D: type 1 diabetes; CGM: continuous glucose monitoring; AUC-ROC: Area Under the Receiver Operating Characteristic Curve; HbA1c: glycated hemoglobin; SES: socioeconomic status; mtmDR: moderate to mild diabetic retinopathy; vtDR: vision-threatening diabetic retinopathy; LDA: linear discriminant analysis: SVM: support vector machine; RF: random forest. mtmDR: more-than-mild diabetic retinopathy; vtDR: vision-threatening diabetic retinopathy.

Skills Programme (CASP) checklists (Supplementary file 3).[40] For the models of the studies included in the review, QUADAS-2 toll has also been added for quality evaluation (Supplementary file 4 single study analysis and Fig. 2 Tabular presentation in the text).[41] This approach ensured that the conclusions were based on high-quality evidence, contributing to more robust and reliable information on the research question.

2.7. Data extraction and synthesis

Data from the selected studies were summarized both in a descriptive table (Supplementary file 5) and narratively. The data extraction process included compiling a specific table (Table 1) containing the following elements: Author and Year, Country, Study Design, Sample (patients with T1D), mean age and gender distribution, timing study and setting,

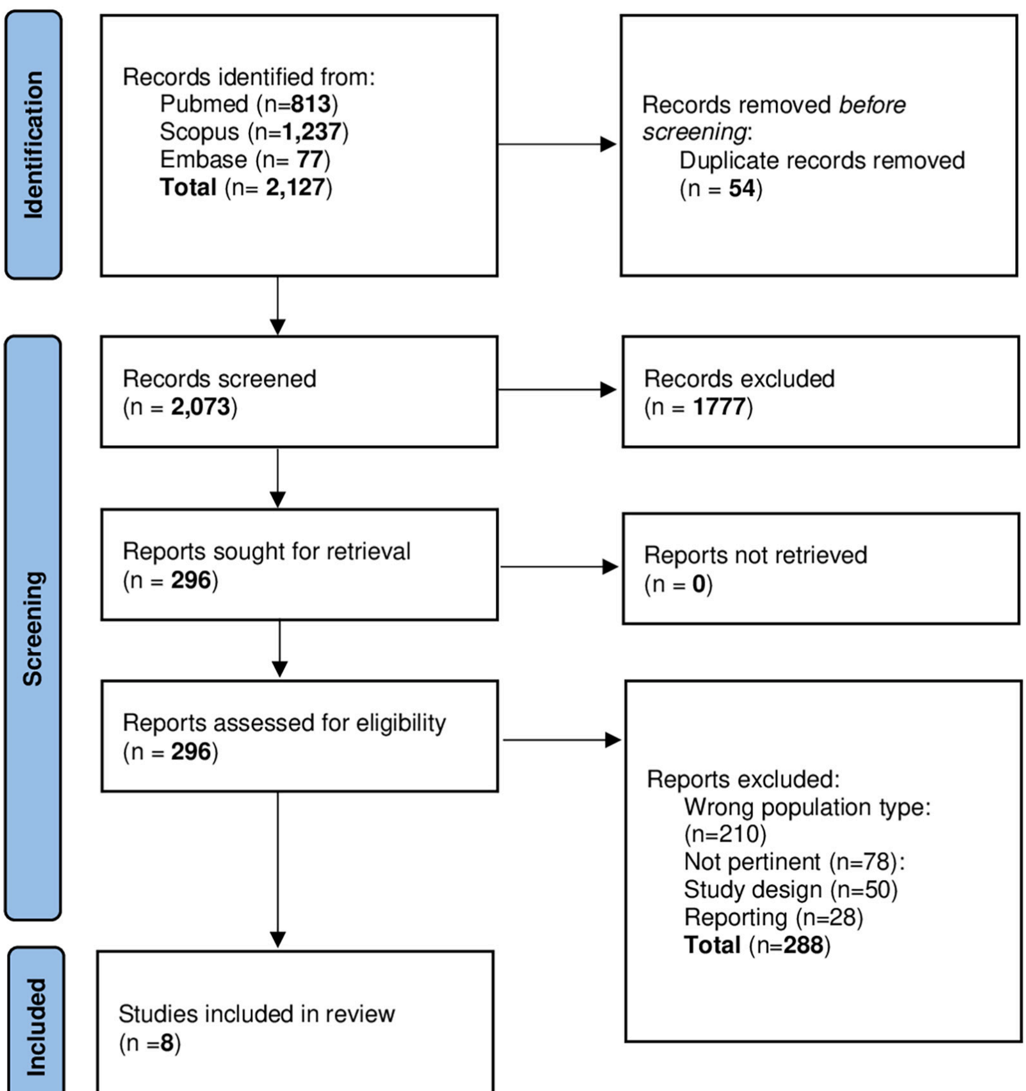

Fig. 1. Flowchart studies included.

sensitivity, specificity, and accuracy, type of AI, specific objective, and results. Additionally, the extracted data were presented as a narrative summary, structured according to the objectives of the review.

3. Results

A total of 2127 articles were identified through the search strategy conducted in the PubMed-Medline, Embase, and Scopus databases. After removing 54 duplicates, 1777 articles were excluded based on title and abstract screening. Subsequently, 296 articles were selected for further evaluation. Of these, *288* were deemed irrelevant (210 for wrong population type and 78 because not pertinent: 50 for not valid study design and 28 for not pertinent reporting adopted). The selection process resulted in the final inclusion of 8 studies (Fig. 1).[42–49]

3.1. Characteristics and synthesis of included studies

The 8 included studies[42–49] were conducted across various countries: four in the USA,[42–44,48] and one each in Taiwan,[47] Australia,[49] Rwanda,[45] and Denmark.[46] Of these, five were experimental or clinical studies,[42,44,45,47,48] while three were observational studies.[43,46,48] The total population across all studies comprised 2717 participants, of whom 1470 had T1D. Three studies[42,48,49] employed autonomous AI systems, two utilized ML models,[43,47] and three implemented DL techniques.[44,45,47] The study designs were uniformly rated as high quality (Tables 1 and Figs. 2 and 3).

3.2. Autonomous AI systems for the identification and screening of DR

In a randomized controlled trial (RCT)[42] conducted in the USA, 163 pediatric patients were analyzed. In the experimental group (EG, n = 81), participants underwent diabetic eye examinations (DED) using an autonomous AI system lasting 5–10 min. In contrast, the control group (CG, n = 82) received educational intervention from an eye care provider (ECP: ophthalmologist or optometrist) on obtaining a DED. In the EG, all 81 subjects (100 %, 95 % CI: 95.5 %, 100 %) (T1D = 59; T2D = 22) completed the DED, compared to only 18 subjects in the CG (22 %, 95 % CI: 12.2 %, 32.4 %) (T1D = 12; T2D = 6). Furthermore, in the EG, 25 out of 81 participants received an "output DED present," and 64 % (16/25) completed a follow-up visit with the ECP, compared to 22 % in the CG. In another diagnostic cross-sectional study[48] conducted across 15 care centers in the USA, 893 subjects were analyzed (T1D = 206). An autonomous AI system (EyeArt system, version 2.1.0) was used to acquire retinal fundus photographs (CFP) at two fields for detecting mild to moderate diabetic retinopathy (mtmDR) and vision-threatening diabetic retinopathy (vtDR). Results were compared to the clinical reference standard of the Early Treatment Diabetic Retinopathy Study (ETDRS) fundus photographs provided by certified personnel (FPRC). The AI system demonstrated high accuracy compared to the reference standard, with a sensitivity for mtmDR of 96 % and specificity of 88 %, and a sensitivity for vtDR of 97 % and specificity of 90 %. Additionally, the AI system successfully classified approximately 97 % of the manually assessed eyes, with most cases not requiring dilation. Finally, in a mixed-methods study[49] conducted in Australia, 203 patients were analyzed (T1D = 70). Initially, an autonomous offline AI system screened for DR and other ocular conditions. Subsequently, in-depth interviews were conducted with healthcare professionals. The area under the curve (AUC), sensitivity, and specificity of the AI system across all clinics were 0.92, 96.9 %, and 87.7 %, respectively. Subgroup analysis revealed higher diagnostic accuracy in patients with T1D (sensitivity: 100 %, specificity: 90.9 %, AUC: 0.96, 95 % CI) compared to those with T2D (sensitivity: 96.4 %, specificity: 85.3 %, AUC: 0.91, 95 % CI). Additionally, 93.7 % (194/207) of participants reported satisfaction or high satisfaction with the AI model, and healthcare staff described the system as easy to use.

BIAS RISK ASSESSMENT				
	Patient Selection	Index test	Reference standard	Flow and timing
Wolf et al. [40]	Low	Low	Unclear	Unclear
Montaser et al. [41]	Unclear	Unclear	Unclear	Unclear
Lee et al. [42]	Low	Low	Low	Low
Mathenge et al. [43]	Low	Low	Low	Low
Al-Sari et al. [44]	Low	Low	Low	Unclear
Lo et al. [45]	Unclear	Unclear	Unclear	Low
Ipp et al. [46]	Low	Low	Low	Unclear
Scheetz et al. [47]	Low	Low	Low	Low

Low	6	6	5	4
High	0	0	0	0
Unclear	1	2	3	4

ASSESSMENT OF APPLICABILITY			
	Patient selection	Index test	Reference standard
Wolf et al. [40]	Low	Low	Low
Montaser et al. [41]	Low	Unclear	Unclear
Lee et al. [42]	Low	Low	Low
Mathenge et al. [43]	Low	Low	Low
Al-Sari et al. [44]	Low	Low	Low
Lo et al. [45]	Low	Unclear	Low
Ipp et al. [46]	Low	Low	Low
Scheetz et al. [47]	Low	Low	Low

Low	8	6	7
High	0	0	0
Unclear	0	2	1

Fig. 2. Tabular presentation of QUADAS-2 (Risk of Bias assessment and Assessments of Applicability).

3.3. ML to identify risks and complications of DR

In an exploratory study[43] conducted in the USA, continuous glucose monitoring (CGM) data from 60 adults with T1D (30 with DR and 30 without DR) were analyzed using three ML models: support vector machine (SVM), linear discriminant analysis (LDA), and random forest (RF). These models were evaluated in two scenarios. In the first scenario, the ML algorithms analyzed all features extracted from the CGM data, with the linear SVM model achieving strong predictive performance (AUC-ROC: 0.90). In the second scenario, glycaemic feature selection was refined using principal component analysis (PCA) prior to ML application, slightly improving performance (AUC-ROC: 0.92). In a

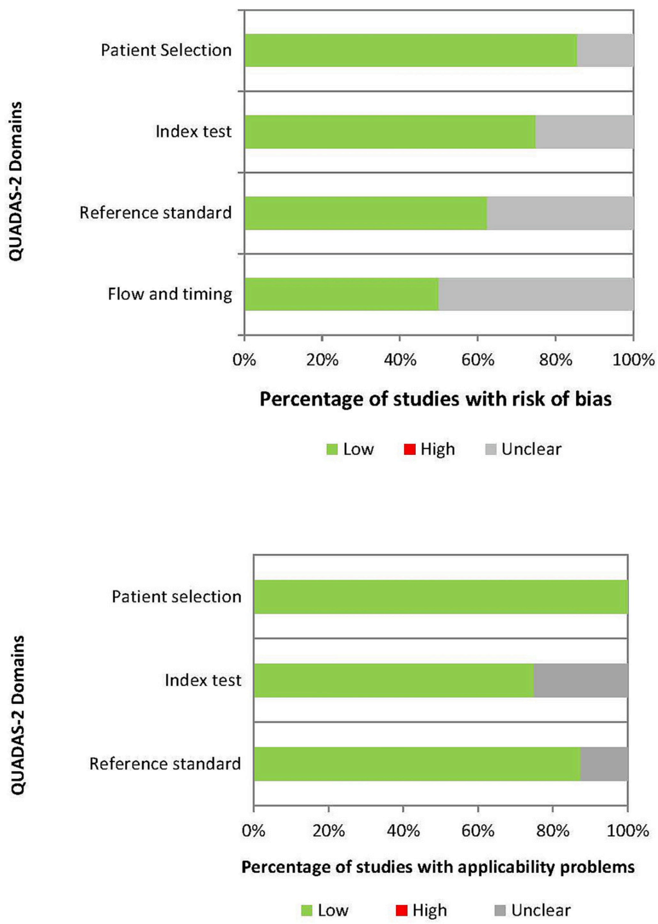

Fig. 3. Tabular presentation of QUADAS-2 (Percentage of studies with risk of bias and Percentage of studies with applicability problems).

retrospective cohort study[46] conducted in Denmark, 383 patients with T1D were analyzed to develop predictive models for DR progression and other diabetes-related complications. Using random forest (RF) algorithms trained on clinical risk factors and omics profiles, the models achieved an AUC-ROC of 0.75 ± 0.14 with clinical data and 0.79 ± 0.16 with molecular measurements. Moreover, the models identified eight biomolecules with strong predictive value for disease progression.

3.4. DL to improve diagnostic accuracy

A non-randomized prospective study[44] conducted in the USA analyzed 265 participants (T1D = 86, 34 %) to assess the feasibility of remote DR screening using the optomap wide-field imaging device (UWF). A DL model developed from these images was compared to clinical evaluations as the reference standard. The model achieved an accuracy of 82.8 % (95 % CI: 80.3–85.2 %), sensitivity of 81.0 % (95 % CI: 78.5–83.6 %), specificity of 73.5 % (95 % CI: 70.6–76.3 %), and an AUC-ROC of 81.0 % (95 % CI: 78.5–83.6 %). In a multi-center RCT[45] in Rwanda, 275 diabetes patients were assessed. The experimental group (EG, n = 136; T1D = 63) underwent DR screening using a DL model with immediate feedback, while the control group (CG, n = 139; T1D = 56) received feedback after 3–5 days from human evaluators. Adherence to follow-up reference exams was significantly higher in the EG (51.5 %) compared to the CG (39.6 %, P = 0.048), marking a 30.1 % increase in adherence. An experimental study[47] in Taiwan analyzed 15,472 retinal images from two datasets using four DL models (Inception-V3, DenseNet-121, VGG16, and Xception). One dataset (EG1, n = 8408) was a public Kaggle dataset with high variability, predominantly from T2D patients, while the other (EG2, n = 7064; T1D = 475) consisted of single-center images exclusively from T1D patients. Models trained on EG1 achieved a mean AUC of 0.74 (±0.03) on the Kaggle dataset and 0.87 (±0.01) on the T1D dataset, with VGG16 performing best (AUC = 0.77). Models trained on EG2 achieved an AUC of 0.88 (±0.03) on the T1D dataset but dropped to 0.57 (±0.02) on the Kaggle dataset, with DenseNet-121 performing best (AUC = 0.91) and VGG16 the worst (AUC = 0.84).

4. Discussion

This SR highlights the growing role of AI in the screening and management of DR among patients with T1D. The findings demonstrate that AI has significant potential to improve DR screening adherence, risk prediction, and diagnostic accuracy, contributing to better disease management and overall healthcare delivery.

The integration of technology into healthcare systems brings numerous benefits for both patients and healthcare professionals in different setting for diabetes care.[50,51] AI, in particular, addresses several pressing challenges in healthcare, improving efficiency and productivity while facilitating high-quality care.[52] Telemedicine, combined with AI, has emerged as a powerful tool to bridge gaps in healthcare delivery, especially for chronic conditions like diabetes. Centralized data management enabled by AI enhances the coordination of care, making treatment plans more personalized and effective.[52] In recent years, various AI applications have been developed to assist clinicians and patients. These include automated reminders for treatment adherence, quality-of-life surveys, and interactive tools such as chatbots that engage patients directly.[53–58] By reducing logistical barriers and streamlining communication, AI promotes better patient engagement and adherence to care protocols. Additionally, AI-driven predictive models enable large-scale deployment, transforming traditional screening strategies and improving the cost-effectiveness of DR management.[53] AI also shows promise in acute and emergency care settings by optimizing resource allocation and improving patient outcomes.[59] For instance, chatbot technologies, including ChatGPT, have been employed to answer patient and healthcare workers queries about chronic diseases, demonstrating improved accessibility and responsiveness.[59–61] Segmentation and detection algorithms further enhance diagnostic accuracy by identifying pathological features in retinal images, such as microaneurysms and hemorrhages, with exceptional precision.[56,62,63] Notably, Ghouali et al. evaluated an AI model that classified patients into urgent, semi-urgent, and non-urgent categories, achieving high discriminant precision.[64] Similarly, Vasudevan et al. reported that computer-assisted diagnostic tools in ophthalmology save time, reduce resource use, and lower costs associated with routine DR screening while maintaining low diagnostic error rates.[54]

One of the most impactful contributions of AI to DR management is its ability to improve screening adherence. For example, autonomous AI systems in experimental studies achieved a 100 % screening completion rate in T1D patients, compared to significantly lower rates in control groups receiving standard care.[42] This increase in adherence is critical for high-risk populations, as regular screening is essential to prevent severe complications like vision-threatening DR. By simplifying the screening process and addressing logistical barriers, AI enhances participation rates and reduces disparities in care delivery. Moreover, ML models have demonstrated remarkable success in predicting DR risks and associated complications. By analyzing CGM data, ML algorithms such as SVM have achieved high predictive accuracy, with AUC-ROC scores exceeding 0.90.[43] These models allow clinicians to identify patients at risk of developing DR early, enabling timely interventions. Additionally, ML models incorporating omics and clinical data have identified key biomarkers associated with DR progression, facilitating personalized care strategies and earlier interventions.[46] Finally, DL models have set new standards for diagnostic accuracy in DR screening. Several studies included in this review reported high sensitivity and specificity for DL-based algorithms when analyzing retinal images. For instance, DL models demonstrated superior performance compared to traditional diagnostic methods in specific scenarios.[44,45,47] Subgroup analyses have further emphasized the importance of tailored approaches, as diagnostic accuracy was higher for T1D patients compared to those with T2D.[49] However, challenges related to the generalizability of DL models persist. Models trained on homogeneous datasets often perform poorly when tested on diverse populations, underscoring the need for datasets that reflect real-world variability.[47] Addressing these limitations will be essential for scaling AI-based solutions globally.

A recent systematic review demonstrated that implementing AI in daily clinical practice could significantly enhance physicians' cognitive abilities and reduce their workload. This is achieved through various automated technologies, including image recognition, speech recognition, and voice recognition, which surpass human capabilities in terms of speed and fatigue resistance.[65,66] For example, AI-based image recognition employs multiple representation layers to analyze each retinal image. However, many AI systems function as black-box models that do not reveal the specific DR lesions they detect (e.g., microaneurysms and retinal hemorrhages).[67] This lack of transparency may limit clinical acceptance and application, emphasizing the need for mechanistic interpretations of AI outputs to facilitate both clinical integration and deeper etiological insights.[68]

In terms of pattern recognition, AI technologies can function like expert physicians by leveraging historical data to make predictions about future outcomes.[65] This capability has the potential to improve the quality of care for DR patients by identifying optimal therapeutic strategies and managing treatment prognoses more effectively. For example, AI can predict post-treatment functional outcomes and the natural progression of DR, providing clinicians with actionable insights.[70] By efficiently generating and processing knowledge from vast datasets, AI systems often outperform even the most experienced experts in terms of accuracy and scalability.[71–73]

To fully realize AI's potential in DR screening and management, future research should focus on improving the generalizability of AI models by training them on diverse datasets, ensuring their applicability across different populations. Furthermore, developing explainable AI systems will foster greater trust and acceptance among clinicians and patients. Finally, implementing robust ethical frameworks that address data privacy, fairness, and equitable access to AI technologies will be critical to ensuring sustainable and impactful deployment.

4.1. Limitations

This review has several limitations that should be acknowledged. One major limitation is the geographical origin of the included studies, as none of the selected studies originated from European countries. Consequently, the findings may not fully capture how AI applications for diabetic retinopathy screening perform in European populations, where healthcare systems, access to technology, and population characteristics may differ. Another limitation is the temporal filter applied during the literature search, which restricted the inclusion of studies to the last five years. While this ensures the review focuses on the most recent advancements, it may have introduced a selection bias by excluding older studies that could provide valuable context. However, it is worth noting that AI applications in this field have seen significant growth only in recent years, making earlier studies less relevant to current technologies. Future research should focus on creating diverse datasets to reduce bias, improving AI transparency for better interpretability, and developing standardized protocols for integration into clinical workflows. Additionally, efforts should be made to ensure equitable access to AI-based DR screening, particularly in underserved areas, to prevent disparities in healthcare outcomes.

4.2. Implications for clinical practice and policymakers

Retinal photography for DR screening is widely regarded as a simple, safe, validated, and acceptable routine diagnostic test. Physicians could integrate these automated tools to reduce workload, assist in diagnostics, and support clinical decision-making. This technology not only facilitates early detection of DR but could also serve as a gateway for identifying other relevant endocrine or systemic diseases, leveraging data derived from AI technologies. The development of automated analysis software using AI-based deep neural learning for retinal images opens new frontiers in personalized medicine. Such tools could be utilized to assess cardiovascular risks in diabetic individuals by analyzing retinal structures and functional changes in the microvasculature.[20] These advancements have the potential to redefine the management of diabetes-related complications in lifestyle view,[74–77] offering precise risk stratification and guiding timely intervention in chronic care in general and with possible predictive extension use.[78–84]

For policymakers, investing in AI-driven retinal imaging technologies could improve the cost-effectiveness of healthcare systems, reduce disparities in access to care, and strengthen preventive strategies against vision-threatening complications and systemic diseases.

5. Conclusion

In conclusion, this review highlights how AI holds immense promise for transforming DR screening and management, particularly for high-risk populations such as T1D patients. By enhancing adherence, risk prediction, and diagnostic accuracy, AI addresses critical gaps in existing care pathways. However, realizing its full potential requires overcoming challenges related to generalizability, transparency, and equitable implementation. With ongoing advancements and a commitment to ethical deployment, AI has the potential to play a pivotal role in reducing the global burden of DR and improving patient outcomes worldwide.

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CRediT authorship contribution statement

Francesco Sacchini: Conceptualization. **Stefano Mancin:** Conceptualization. **Giovanni Cangelosi:** Conceptualization. **Sara Morales Palomares:** Writing – original draft. **Gabriele Caggianelli:** Validation. **Francesco Gravante:** Writing – review & editing. **Fabio Petrelli:** Supervision.

Review protocol registration

The protocol was registered on Open Science Framework (OSF) Register doi.org/10.17605/OSF.IO/TJ9UH.

Ethical statement

All prior approvals were requested and the study adhered to the guidelines of the Declaration of Helsinki and received approval from the Institutional Review Board of Humanitas University Milan. All participants involved have expressed informed consent to the processing of personal data for scientific purposes.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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